SAÉ 5 ROM.03 - Déploiement de Services Multimédias et Intégration Web

Objectif : Déployer un serveur de téléphonie Asterisk et l'intégrer dans un portail Web de visioconférence.

Environnement : Virtualisation sous Alpine Linux, VLAN 800, terminaux IP physiques (Fanvil, Cisco).

INFRASTRUCTURE ET TÉLÉPHONIE PHYSIQUE

Déploiement d'Asterisk

Déploiement réussi

d'Asterisk sur une VM Alpine

Linux.

```
rt-mv:~# service asterisk status
 * status: started
rt-mv:~# _
```

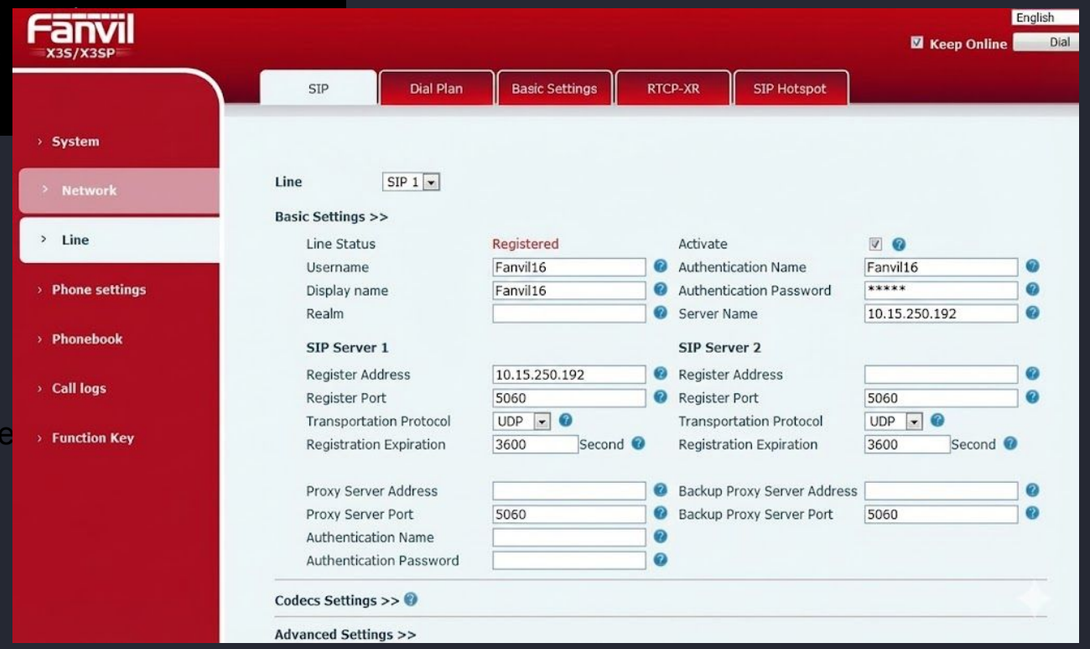
INFRASTRUCTURE ET TÉLÉPHONIE PHYSIQUE

Configuration des terminaux

Configuration et enregistrement de terminaux hétérogènes :

- Fanvil : Configuration via interface Web.
- Cisco : Mise en place d'un serveur TFTP pour le provisionnement des fichiers de configuration.

```
Endpoint:  301                                                  Not in use    0 of inf
    InAuth:  301-auth/301
       Aor:  301                                                1
   Contact:  301/sip:ln5u6023@10.31.25.96:37618;transpo c33c4f0b8c NonQual         nan
 Transport:  transport-ws              wss      0      0  0.0.0.0:8089

Endpoint:  302                                                  Not in use    0 of inf
    InAuth:  302-auth/302
       Aor:  302                                                1
   Contact:  302/sip:oinku7eu@10.16.20.193:51830;transp a765d330bb NonQual         nan
 Transport:  transport-ws              wss      0      0  0.0.0.0:8089

Endpoint:  Cisco16                                              Not in use    0 of inf
    InAuth:  auth-Cisco16/Cisco16
       Aor:  Cisco16                                            1
   Contact:  Cisco16/sip:Cisco16@10.17.1.36:5060;transp 220e898534 NonQual         nan
 Transport:  transport-udp             udp      0      0  0.0.0.0:5060

Endpoint:  Fanvil16                                             Not in use    0 of inf
    InAuth:  auth-Fanvil16/Fanvil16
       Aor:  Fanvil16
   Contact:  Fanvil16/sip:Fanvil16@10.17.1.16:5287
 Transport:  transport-udp             udp      0      0

Objects found: 4
```



Configuration via le fichie
créer les utilisateurs.

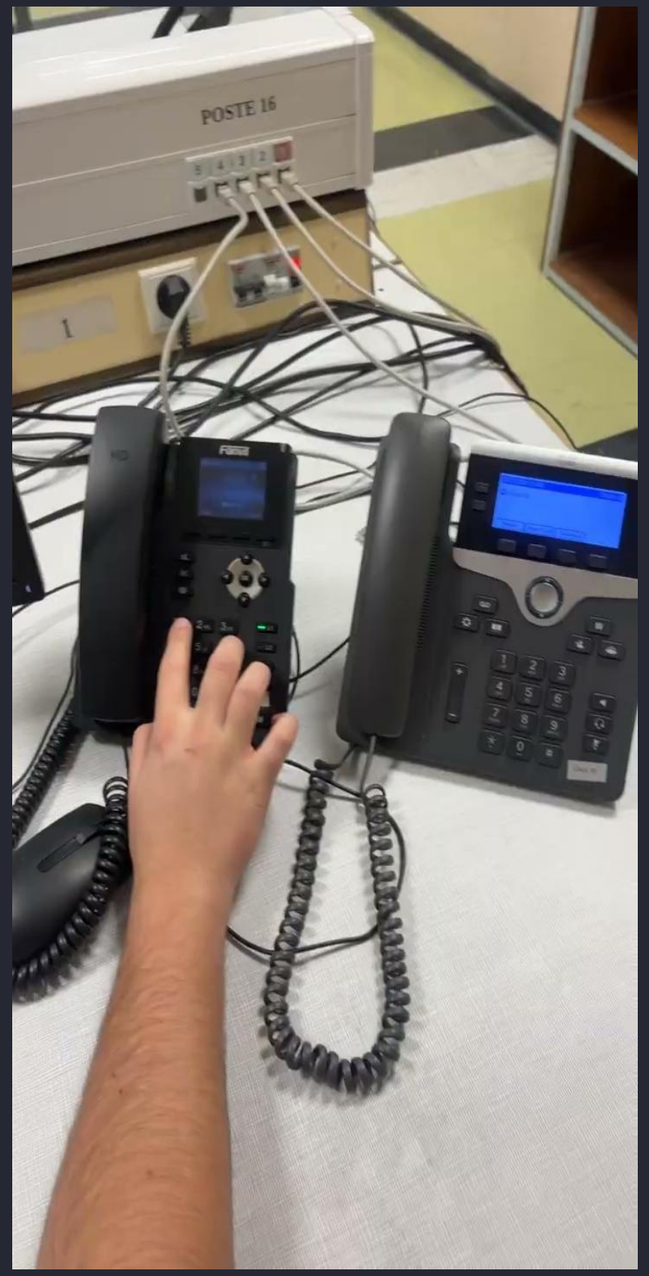
INFRASTRUCTURE ET TÉLÉPHONIE PHYSIQUE

Validation

Mise en place du plan de numérotation, de la messagerie vocale et validation des appels par analyse Wireshark.

```
[internal]
exten=>1601,1,Dial(PJSIP/Fanvil16,12)
same=>n,VoiceMail(1601@default,u)
same=>n,Hangup()
exten=>1602,1,Dial(PJSIP/Cisco16,12)
same=>n,VoiceMail(1602@default,u)
same=>n,Hangup()

exten=>500,1,VoiceMailMain()
same=>n,Hangup()

exten => 301,1,Dial(PJSIP/301,20)
exten => 302,1,Dial(PJSIP/302,20)

exten => _3XX,1,Dial(PJSIP/${EXTEN},20)
same => n,Answer()
same => n,Hangup()

exten => _XXXX,1,Dial(PJSIP/${EXTEN})
```

Appel entre deux téléphones



Capture Wireshark de l'appel

2261	36.877383032	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2262	36.891709743	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2263	36.904626952	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2264	36.904680116	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2265	36.912053947	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2266	36.932169991	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2267	36.936560617	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2268	36.952055140	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2269	36.963339167	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2270	36.972196285	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2271	36.989935078	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2272	36.991937900	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2273	37.012171171	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2274	37.015279263	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2275	37.032295108	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2276	37.043578979	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2277	37.049012364	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2278	37.052137812	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2279	37.055824088	10.15.250.192	10.16.20.193	UDP	156	13356 → 39517 Len=114
2280	37.056441926	10.15.250.192	10.16.20.193	TCP	786	5038 → 41178 [PSH, ACK] Seq=14067 Ack=1 Win=510 Len=720 TSval=2716788155 TSecr=…
2281	37.056487158	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=14787 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2282	37.056442072	10.15.250.192	10.16.20.193	TCP	786	5038 → 41164 [PSH, ACK] Seq=14067 Ack=1 Win=510 Len=720 TSval=2716788155 TSecr=…
2283	37.056546557	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=14787 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2284	37.056702558	10.15.250.192	10.16.20.193	TCP	785	5038 → 41178 [PSH, ACK] Seq=14787 Ack=1 Win=510 Len=719 TSval=2716788155 TSecr=…
2285	37.056710093	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=15506 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2286	37.056702643	10.15.250.192	10.16.20.193	TCP	785	5038 → 41164 [PSH, ACK] Seq=14787 Ack=1 Win=510 Len=719 TSval=2716788155 TSecr=…
2287	37.056740945	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=15506 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2288	37.072193713	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182

Messagerie Vocale

DÉVELOPPEMENT DU PORTAIL WEB

- Création de l'interface graphique du portail de visioconférence.
- Intégration fonctionnelle d'un service de messagerie instantanée (Chat).

- Note : L'interface est prête pour la vidéo, mais les flux temps réel restent dépendants de la résolution des problèmes de certificats sécurité.

Validation de l'enregistrement des Endpoints PJSIP

```
MariaDB [softphone_chat]> select * from utilisateurs;
+----+----------+------------------------------------------------------------+-----------+
| id | username | password_hash                                              | extension |
+----+----------+------------------------------------------------------------+-----------+
|  1 | baptiste | ea3a05a5ab9792896896a275a46e998504c464005717af928d7dd7e330bef5f3 |       301 |
|  2 | fanvil   | ea3a05a5ab9792896896a275a46e998504c464005717af928d7dd7e330bef5f3 |      1601 |
|  3 | roman,   | ea3a05a5ab9792896896a275a46e998504c464005717af928d7dd7e330bef5f3 |       302 |
|  4 | cisco    | ea3a05a5ab9792896896a275a46e998504c464005717af928d7dd7e330bef5f3 |      1602 |
+----+----------+------------------------------------------------------------+-----------+
4 rows in set (0,000 sec)

MariaDB [softphone_chat]>
```

AMI

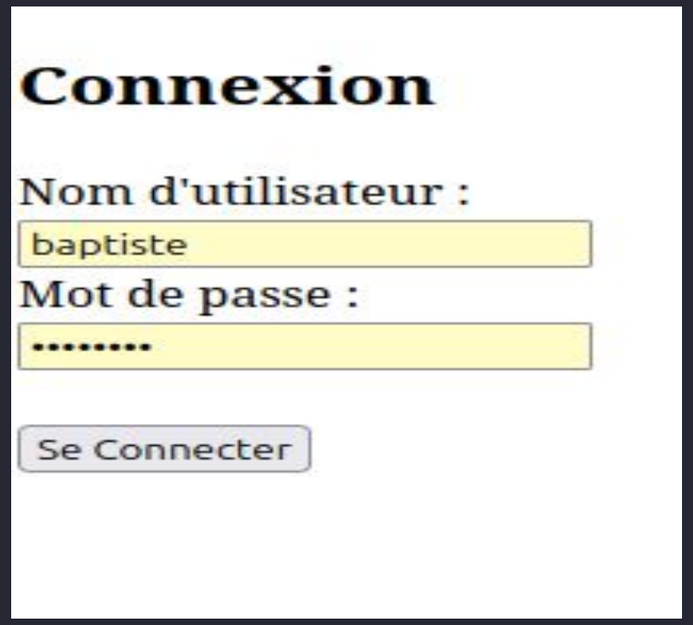
Flask : app.py

```
AMI_HOST =
'10.15.250.192'
AMI_PORT = 5038
AMI_USER = 'webapp'
AMI_SECRET = '1234'
AMI_CONTEXT = 'internal'
```

```
[webapp]
secret=1234
permit=10.16.20.193/255.255.255.255
read=system,call,log,verbose,command,agent,user,config,dtmf,reporting,cdr,diaplan,originate
write=system,call,log,verbose,command,agent,user,config,dtmf,reporting,cdr,diaplan,originate
```

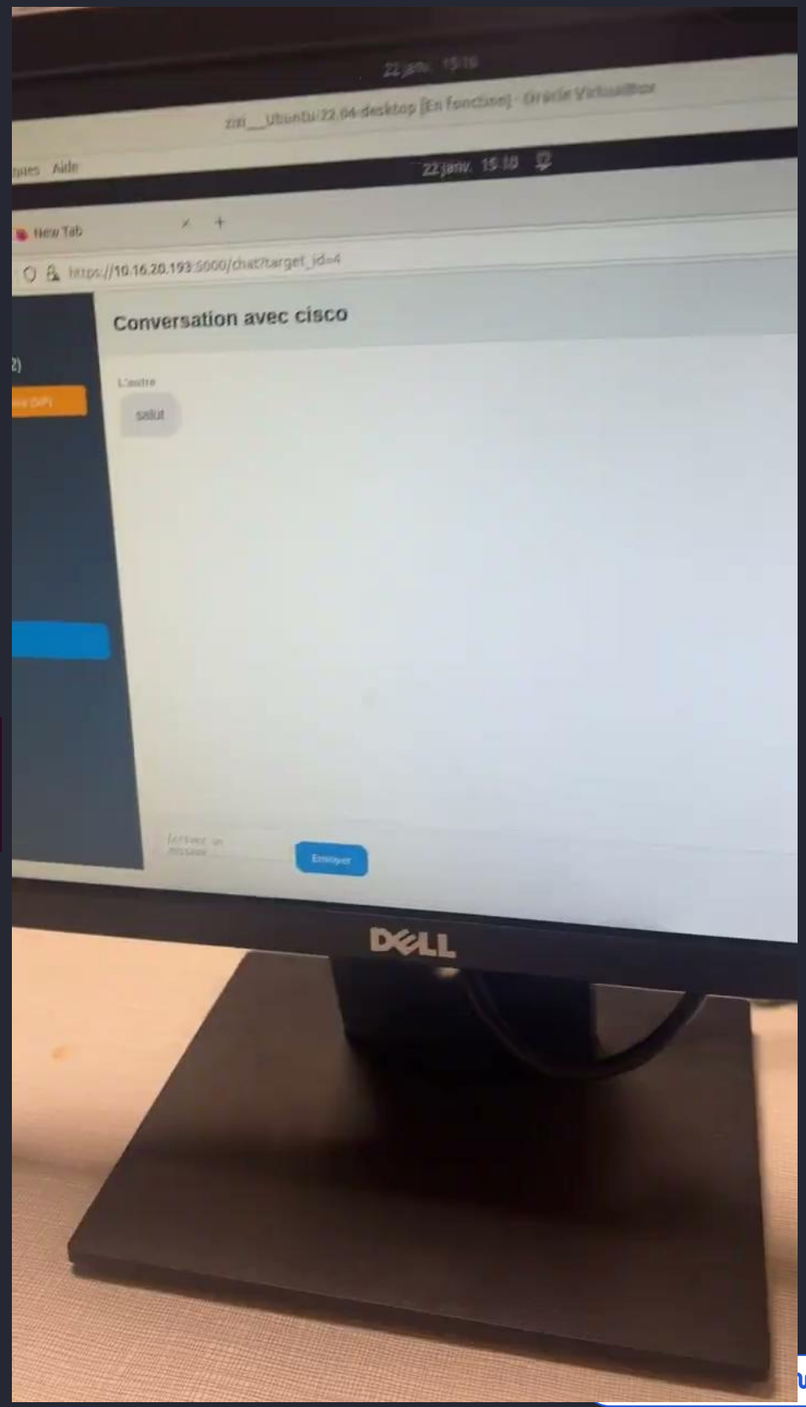
Manager.conf

Interface Web



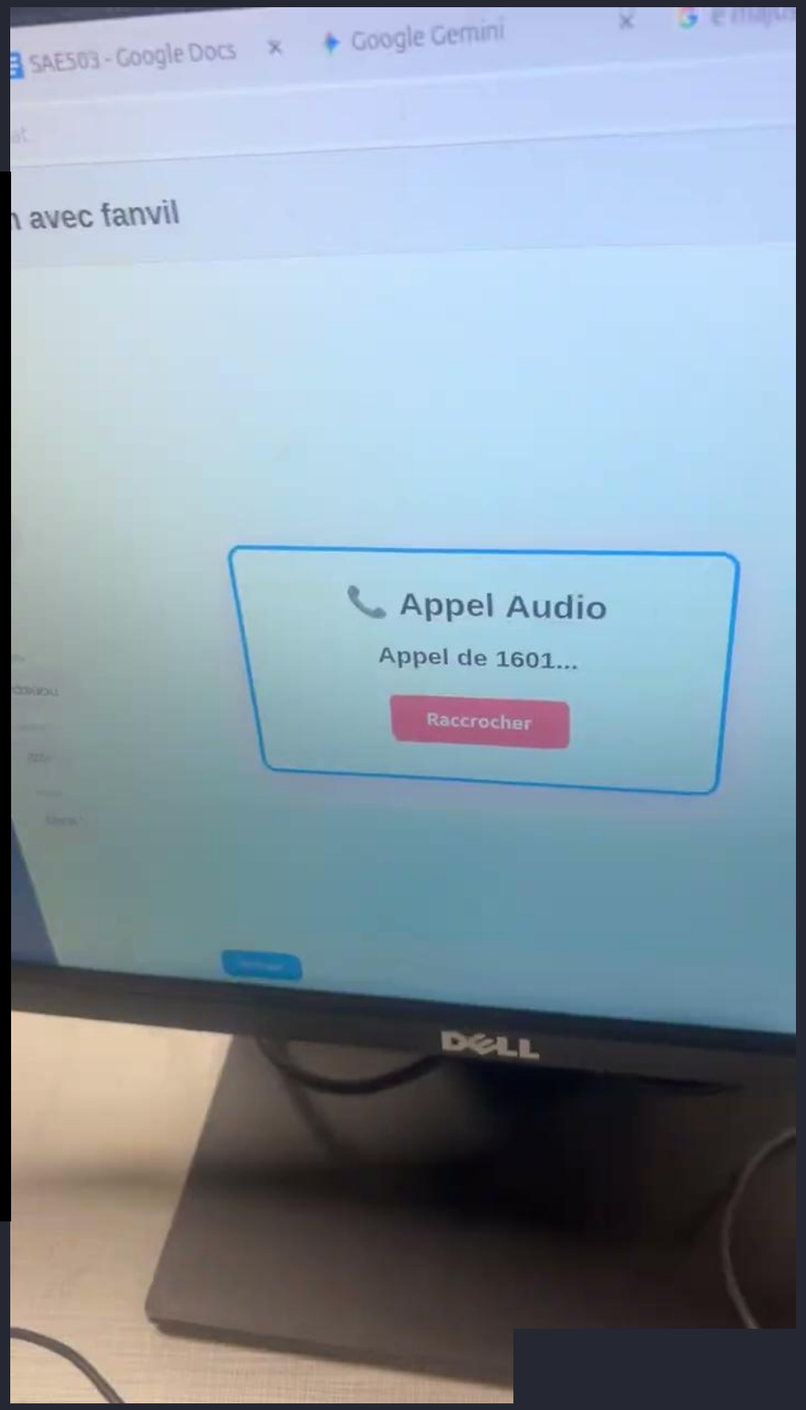
Echanges de chat

36	4	1	salut	2026-01-22 14:10:04
37	4	3	salut	2026-01-22 14:10:22
38	3	4	coucou	2026-01-22 14:10:34
39	4	3	ça va?	2026-01-22 14:10:44



Appel Web vers Téléphone

```
<--- Received SIP request (476 bytes) from WSS:10.16.20.193:34480 --->
BYE sip:10.15.250.192:8089;transport=ws SIP/2.0
Via: SIP/2.0/WSS 85rdcvj7kkmt.invalid;branch=z9hG4bK949395
Max-Forwards: 69
To: <sip:1601@10.15.250.192>;tag=INepBMEIOeQyfkb-1D5KAIxcOWIJE4gL
From: "301" <sip:301@10.15.250.192>;tag=0h9e0dosru
Call-ID: tafeahq15mcgr0hrmlhj
CSeq: 6514 BYE
Reason: SIP ;cause=500; text="Dialog Error"
Allow: INVITE,ACK,CANCEL,BYE,UPDATE,MESSAGE,OPTIONS,REFER,INFO,NOTIFY
Supported: outbound
User-Agent: JsSIP 3.9.2
Content-Length: 0


<--- Transmitting SIP response (350 bytes) to WSS:10.16.20.193:34480 --->
SIP/2.0 481 Call/Transaction Does Not Exist
Via: SIP/2.0/WSS 85rdcvj7kkmt.invalid;rport;received=10.16.20.193;branch=z9hG4bK949395
Call-ID: tafeahq15mcgr0hrmlhj
From: "301" <sip:301@10.15.250.192>;tag=0h9e0dosru
To: <sip:1601@10.15.250.192>;tag=INepBMEIOeQyfkb-1D5KAIxcOWIJE4gL
CSeq: 6514 BYE
Server: Asterisk PBX 18.2.2
Content-Length:  0


*t-mv*CLI> _
```



162	20.215450742	10.16.20.193	10.15.250.192	TLSv1.2	96	Application Data
163	20.215487320	10.16.20.193	10.15.250.192	TLSv1.2	1922	Application Data
164	20.216927120	10.15.250.192	10.16.20.193	TCP	66	8089 → 52110 [ACK] Seq=29 Ack=1919 Win=496 Len=0 TSval=2716715723 TSecr=106…
165	20.217440707	10.15.250.192	10.16.20.193	TLSv1.2	561	Application Data
166	20.217456464	10.16.20.193	10.15.250.192	TCP	66	52110 → 8089 [ACK] Seq=1919 Ack=524 Win=501 Len=0 TSval=1061870237 TSecr=27…
167	20.221961846	10.16.20.193	10.15.250.192	TLSv1.2	497	Application Data
168	20.225035930	10.16.20.193	10.15.250.192	TLSv1.2	96	Application Data
169	20.225069385	10.16.20.193	10.15.250.192	TLSv1.2	2194	Application Data
170	20.225881568	10.15.250.192	10.16.20.193	TCP	66	8089 → 52110 [ACK] Seq=524 Ack=2380 Win=501 Len=0 TSval=2716715733 TSecr=10…
171	20.226838825	10.15.250.192	10.16.20.193	TCP	66	8089 → 52110 [ACK] Seq=524 Ack=4508 Win=501 Len=0 TSval=2716715734 TSecr=10…
172	20.227996696	10.15.250.192	10.16.20.193	TLSv1.2	390	Application Data
173	20.230032392	10.15.250.192	10.16.20.193	TCP	398	5038 → 41178 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=332 TSval=2716715737 TSecr=…
174	20.230058734	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=333 Win=501 Len=0 TSval=1061870250 TSecr=27167…
175	20.230032723	10.15.250.192	10.16.20.193	TCP	398	5038 → 41164 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=332 TSval=2716715737 TSecr=…
176	20.230082452	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=333 Win=501 Len=0 TSval=1061870250 TSecr=27167…
177	20.233152787	10.15.250.192	10.16.20.193	TCP	411	5038 → 41178 [PSH, ACK] Seq=333 Ack=1 Win=510 Len=345 TSval=2716715740 TSec…
178	20.233176790	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=678 Win=501 Len=0 TSval=1061870253 TSecr=27167…
179	20.233153027	10.15.250.192	10.16.20.193	TCP	411	5038 → 41164 [PSH, ACK] Seq=333 Ack=1 Win=510 Len=345 TSval=2716715740 TSec…
180	20.233211151	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=678 Win=501 Len=0 TSval=1061870253 TSecr=27167…
181	20.233895093	10.15.250.192	10.16.20.193	TCP	455	5038 → 41164 [PSH, ACK] Seq=678 Ack=1 Win=510 Len=389 TSval=2716715741 TSec…
182	20.233906994	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=1067 Win=501 Len=0 TSval=1061870253 TSecr=2716…
183	20.233895183	10.15.250.192	10.16.20.193	TCP	455	5038 → 41178 [PSH, ACK] Seq=678 Ack=1 Win=510 Len=389 TSval=2716715741 TSec…
184	20.233929459	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=1067 Win=501 Len=0 TSval=1061870253 TSecr=2716…
185	20.234038709	10.15.250.192	10.16.20.193	TCP	403	5038 → 41178 [PSH, ACK] Seq=1067 Ack=1 Win=510 Len=337 TSval=2716715741 TSe…
2261	36.877383032	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2262	36.891709743	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2263	36.904626952	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2264	36.904680116	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2265	36.912053947	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2266	36.932169991	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2267	36.936560617	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2268	36.952055140	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2269	36.963339167	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2270	36.972196285	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2271	36.989935078	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2272	36.991937900	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2273	37.012171171	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2274	37.015279263	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2275	37.032295108	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2276	37.043578979	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2277	37.049012364	10.16.20.193	10.15.250.192	UDP	224	39517 → 13356 Len=182
2278	37.052137812	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182
2279	37.055824088	10.15.250.192	10.16.20.193	UDP	156	13356 → 39517 Len=114
2280	37.056441926	10.15.250.192	10.16.20.193	TCP	786	5038 → 41178 [PSH, ACK] Seq=14067 Ack=1 Win=510 Len=720 TSval=2716788155 TSecr=…
2281	37.056487158	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=14787 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2282	37.056442072	10.15.250.192	10.16.20.193	TCP	786	5038 → 41164 [PSH, ACK] Seq=14067 Ack=1 Win=510 Len=720 TSval=2716788155 TSecr=…
2283	37.056546557	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=14787 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2284	37.056702558	10.15.250.192	10.16.20.193	TCP	785	5038 → 41178 [PSH, ACK] Seq=14787 Ack=1 Win=510 Len=719 TSval=2716788155 TSecr=…
2285	37.056710093	10.16.20.193	10.15.250.192	TCP	66	41178 → 5038 [ACK] Seq=1 Ack=15506 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2286	37.056702643	10.15.250.192	10.16.20.193	TCP	785	5038 → 41164 [PSH, ACK] Seq=14787 Ack=1 Win=510 Len=719 TSval=2716788155 TSecr=…
2287	37.056740945	10.16.20.193	10.15.250.192	TCP	66	41164 → 5038 [ACK] Seq=1 Ack=15506 Win=501 Len=0 TSval=1061942704 TSecr=2716788…
2288	37.072193713	10.15.250.192	10.16.20.193	UDP	224	13356 → 39517 Len=182

Conclusion et Bilan du Projet

Objectifs Atteints

- Administration Système & VoIP
- Gestion de l'hétérogénéité
- Architecture de données

Gestion des problématiques de sécurité

- Confrontation au Web moderne
- Analyse de flux
- Résilience

Bilan Personnel & Professionnel

- Adaptabilité
- Esprit critique